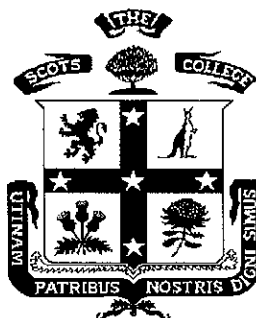


THE SCOTS COLLEGE



YEAR 12 MATHEMATICS EXTENSION 2

HSC TRIAL

AUGUST 2008

General Instructions

- All questions are of equal value
- Working time - 3 hours + 5 minutes reading time.
- Write using blue or black pen
- Board approved calculators may be used
- Start a new booklet for each question
- All necessary working should be shown in every question
- A Standard Integrals Table is attached

TOTAL MARKS: 120

WEIGHTING: 40 %

QUESTION 1 [15 MARKS]**MARKS****a.** If $z = 1 + i$, find:

(i) $|z|$ [1]

(ii) $\arg z$ [1]

(iii) z^{-6} in the form $x + iy$ [2]

b. Solve the equation for z [3]

$$z\bar{z} + 2iz = 12 + 6i$$

c. What is the locus in the Argand Diagram of the point Z which represents the complex number z where: [2]

$$z\bar{z} - 2(z + \bar{z}) = 5$$

d. The origin and the points representing the complex numbers z , $\frac{1}{z}$ and $z + \frac{1}{z}$ are joined to form a quadrilateral. Write down the conditions for z so that the quadrilateral will be a:

(i) rhombus [1]

(ii) square [2]

e. Prove by induction that, for all integers $n \geq 1$, [3]

$$(\cos \theta - i \sin \theta)^n = \cos(n\theta) - i \sin(n\theta)$$

- a.** If $f(x) = (x+2)(x-1)$, sketch the graphs of the following functions on separate diagrams.

(i) $y = f(x)$ [1]

(ii) $y = \frac{1}{|f(x)|}$ [2]

(iii) $y = \log_e(f(x))$ [2]

(iv) $y^2 = f(x)$ [2]

- b.** **(i)** By using implicit differentiation, state where $\frac{dy}{dx}$ is undefined for $y^2 = -x^2(x+2)(x-1)$. [2]

- (ii)** Hence or otherwise, sketch the curve. [2]

c. Let $f(x) = x - 2 + \frac{3}{x+2}$

- (i)** Find the points for which $f(x) = 0$. [1]

- (ii)** Find the asymptotes. [2]

- (iii)** Sketch the curve. Show **all** asymptotes and the x and y intercepts. (There is no need to find or label stationary points.) [1]

a. Evaluate $\int_0^{\frac{\pi}{4}} x \sin 2x \, dx$ [3]

b. Find $\int x\sqrt{1-x} \, dx$ [2]

c. Find $\int \frac{1}{x(1+x^2)} \, dx$ [3]

d. By completing the square and using the table of Standard Integrals, find $\int \frac{dx}{\sqrt{x^2 - 4x + 1}}$ [2]

e. Explain why the following integral cannot be evaluated. [1]

$$\int_0^5 \frac{1}{3-x} \, dx$$

f. Evaluate $\int_0^{\pi/3} \frac{\tan x}{1 + \cos x} \, dx$, using the substitution $t = \tan \frac{x}{2}$. [4]

a. For the ellipse $x^2 + 4y^2 = 100$

(i) Write down the eccentricity, the co-ordinates of the foci and the equations of the directrices. [3]

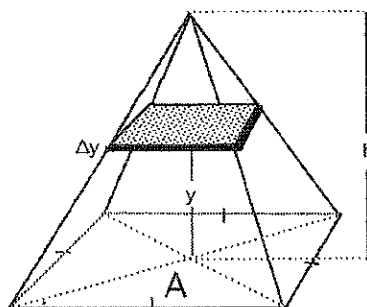
(ii) Sketch the graph of the ellipse showing the above features. [1]

(iii) Find the equation of the tangent and normal to the ellipse at the point $P(8,3)$. [3]

(iv) The normal at P meets the major axis at G . A point K lies on the tangent which passes through the point $P(8,3)$. A perpendicular from K passes through the origin O . Prove that $PG \times OK$ is equal to the square of the length of the semi-minor axis. [2]

b. A particle is projected from a point on a straight line with velocity $u \text{ ms}^{-1}$ and moves in such a way that when it has travelled a distance of x metres it has a velocity of $v = \frac{u}{4+ux} \text{ ms}^{-1}$. Prove that the acceleration of the particle is $-v^3 \text{ ms}^{-2}$. [2]

c. One of the largest pyramids in Egypt is approximately 150m high and has a square base with a base area of approximately 50,000m². The diagram below shows a square based pyramid with a base area A and height h . The thickness of the cross section at height y is Δy .



(i) Show that the area of the cross section at height y can be represented as:

$$A \times \left(\frac{h-y}{h} \right)^2 \quad [1]$$

(ii) Find the volume of the pyramid by using the slicing technique. [3]

a. Let $I_n = \int_0^\pi x^n \sin x \, dx$, where n is a positive integer.

(i) Show that $I_n = \pi^n - n(n-1)I_{n-2}$, for $n \geq 2$ [3]

(ii) Hence evaluate I_5 [3]

b. Find, by the method of cylindrical shells, the volume of the solid generated when the region bounded by the curve $y = x^2 + 1$, the line $x = 2$ and the co-ordinate axes is rotated about the line $x = 3$. [5]

c. Let θ be a real number and consider $(\cos \theta + i \sin \theta)^3$

(i) Prove $\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$ [3]

(ii) Find a similar expression for $\sin 3\theta$ [1]

- a. Let α, β, δ be the roots of the equation $x^3 + qx + r = 0$, where q and r are integers. Write down, in terms of q and r , the cubic equation whose roots are:

(i) $\alpha^{-1}, \beta^{-1}, \delta^{-1}$ [2]

(ii) $\alpha^2, \beta^2, \delta^2$ [2]

- b. Consider the following statements about a polynomial $P(x)$.

Indicate whether each of the following statements is true or false. Give reasons for your answer.

(i) If $P(x)$ is even, then $P'(x)$ is odd. [2]

(ii) If $P'(x)$ is even, then $P(x)$ is odd. [2]

c. (i) Evaluate $\int_0^1 \frac{1}{\sqrt{1-x^2}} dx$ [2]

(ii) Show that for $n \geq 2$ and $0 \leq x \leq \frac{1}{2}$, then $1 \geq 1-x^n \geq 1-x^2$ [2]

(iii) If $n \geq 2$, explain carefully why $\frac{1}{2} \leq \int_0^1 \frac{dx}{\sqrt{1-x^n}} \leq \frac{\pi}{6}$ [3]

- a.** Consider a sequence of numbers $a_1, a_2, a_3 \dots$ where $a_1 = 2, a_2 = 3$ and $a_n = 3a_{n-1} - 2a_{n-2}$ for all $n \geq 3$. Use mathematical induction to prove that $a_n = 2^{n-1} + 1$ for all $n \geq 1$. [5]
- b.** A particle is projected from a height H above a horizontal plane with speed V at an angle of elevation θ to the horizontal.
- (i)** If the range of the particle in the horizontal plane is R , show that $gR^2 \sec^2 \theta = 2V^2(R \tan \theta + H)$. [4]
- (ii)** If R_1 is the maximum value of R and θ_1 is the corresponding value of θ , prove that $R_1 = \frac{v}{g} \sqrt{v^2 + 2gH}$ and $\theta_1 = \tan^{-1} \left(\frac{v^2}{gR} \right)$ [4]
- (iii)** Show that $\tan 2\theta_1 = \frac{R_1}{H}$ [2]

QUESTION 8**[15 MARKS]****START A NEW BOOKLET**

a. Let $m = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$

(i) Prove that $1 + m + m^2 + \dots + m^6 = 0$ **[2]**

(ii) The complex number $\alpha = m + m^2 + m^4$ is a root of the quadratic equation $x^2 + ax + b = 0$, where a and b are real. The second root of the quadratic equation $x^2 + ax + b = 0$ is β . Express β in terms of positive powers of m . Justify your answer. **[2]**

(iii) Find the values of the coefficients a and b . **[2]**

(iv) Deduce that $\sin \frac{2\pi}{7} + \sin \frac{3\pi}{7} - \sin \frac{\pi}{7} = \frac{\sqrt{7}}{2}$ **[2]**

b. Given that $p + q + r = 1$ and $p + q + r \geq 3\sqrt[3]{pqr}$ (where p, q, r are positive real numbers):

(i) Prove that $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} \geq 9$ **[4]**

(ii) Hence, or otherwise, show $\left(\frac{1}{p} - 1\right)\left(\frac{1}{q} - 1\right)\left(\frac{1}{r} - 1\right) \geq 8$ **[3]**

END OF EXAMINATION

Question 1

Solutions Scots 08 Trial

i) $|z| = \sqrt{2}$ ✓ ii) $\arg z = \frac{\pi}{4}$ ✓ iii) $z = [\sqrt{2} \operatorname{cis}(\frac{\pi}{4})]$
 $z^{-6} = \frac{1}{8} \operatorname{cis}(-\frac{3\pi}{2})$ ✓
 $= +\frac{1}{8} i$ ✓

b) $z\bar{z} + 2iz = 12 + 6i$, let $z = x + iy$
 $(x+iy)(x-iy) + 2i(x+iy) = 12 + 6i$
 $x^2 + y^2 + 2ix - 2y = 12 + 6i$ ✓

$2x = 6$ (equating imaginary parts)
 $x = 3$ ✓

$\therefore x^2 + y^2 - 2y = 12$

$y^2 - 2y - 3 = 0$ ✓

$(y-3)(y+1) = 0 \therefore y = 3 \text{ or } -1$

$\therefore z = 3 + 3i \text{ or } z = 3 - i$

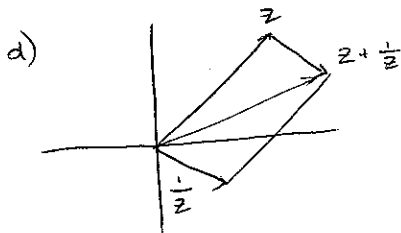
c) $z\bar{z} - 2(z + \bar{z}) = 5$, let $z = x + iy$

$x^2 + y^2 - 2(2x) = 5$

$x^2 - 4x + y^2 = 5$ ✓

$(x-2)^2 + y^2 = 9$ ✓

$\therefore$ locus is a circle, centre 2, radius 3.



i) z must have $|z| = 1$

ii) z must have $|z| = 1$
 and $\arg z = \frac{\pi}{4}$ or $-\frac{\pi}{4}$
 $(\frac{3\pi}{4} \text{ or } -\frac{3\pi}{4})$

i) $|z| = |\frac{1}{z}|$

ii) $z = \frac{1}{z} \therefore |z + \frac{1}{z}| = |z - \frac{1}{z}|$

✓x

Q1 cont...

e) when $n=1$, LHS=RHS

assume true for $n=k$

$(\cos \theta - i \sin \theta)^k = \cos(k\theta) - i \sin(k\theta)$

prove true for $n=k+1$

$(\cos \theta - i \sin \theta)^k (\cos \theta - i \sin \theta)$

$(\cos k\theta - i \sin k\theta)(\cos \theta - i \sin \theta)$ by assumption ✓

$\cos k\theta \cos \theta - i \sin \theta \cos k\theta - i \sin k\theta \cos \theta - \sin \theta \sin k\theta$

$\cos k\theta \cos \theta - \sin \theta \sin k\theta - i(\sin k\theta \cos \theta + \cos k\theta \sin \theta)$ ✓

$\cos(k\theta + \theta) - i(\sin(k\theta + \theta))$

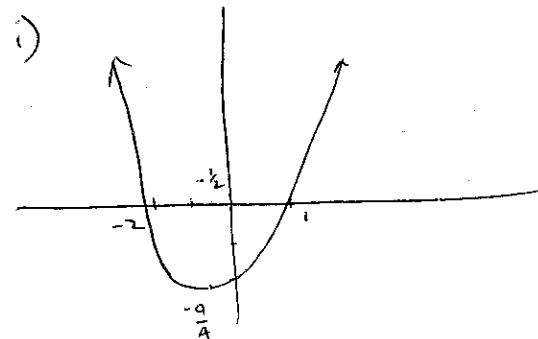
$\cos(k+1)\theta - i \sin(k+1)\theta$ as required. ✓

If result true for $n=k$, then true for $n=k+1$. Since true for $n=1$, it is true for $n=1+1=2$ and so on. Hence true for all positive integers.

Question 2

⊗ Note:

a) i)



Q62

$$-x^2(x^2+x-2) \\ = -x^4 - x^3 + 2x^2$$

b) i) $y^2 = -x^2(x+2)(x-1)$

$$y^2 = (-x^3 + 2x^2)(x-1)$$

$$= -x^4 + x^3 - 2x^3 + 2x^2$$

$$= -x^4 + x^3 + 2x^2$$

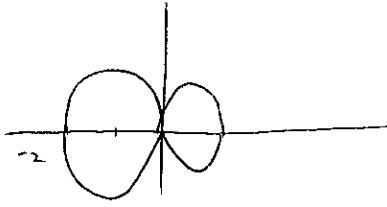
$$2y \cdot \frac{dy}{dx} = -4x^3 + 3x^2 + 4x$$

$$\frac{dy}{dx} = \frac{-4x^3 + 3x^2 + 4x}{2y}$$

$\frac{dy}{dx}$ is undefined for $y=0$

ie, $x=0, 1, -2$

ii)



vertical tangents at $x=-1, 2$

tangent undefined at $x=0$

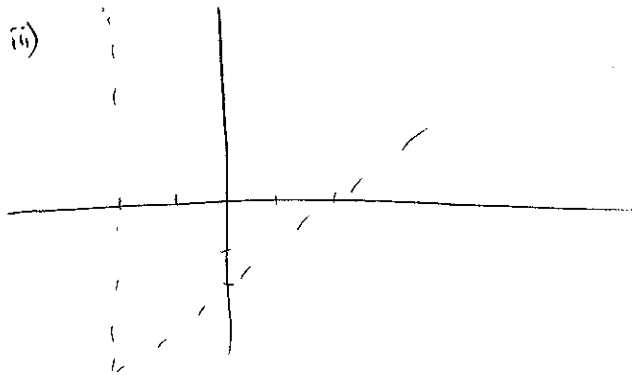
c) i) $f(x) = x-2 + \frac{3}{x+2}$

$$= \frac{x^2 - 4 + 3}{x+2} = \frac{x^2 - 1}{x+2} = 0 \text{ when } x = \pm 1 \quad \checkmark$$

ii) $x=-2$ (vertical asymptote) $\checkmark$

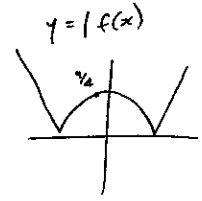
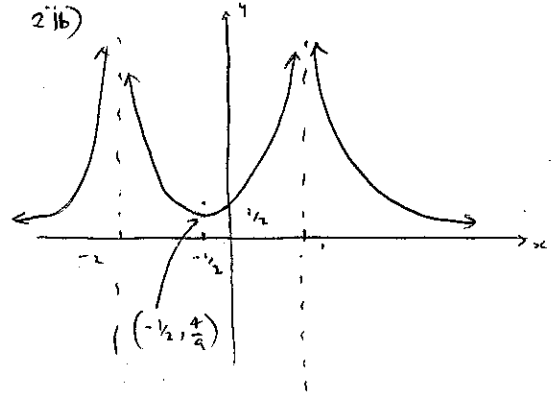
as $x \rightarrow \pm \infty$, $y \rightarrow x-2$ $\therefore y=x-2$ (oblique asymptote) $\checkmark$

iii)

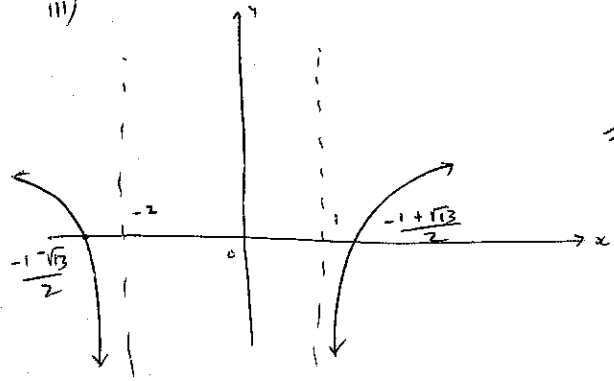


p70.

2) b)



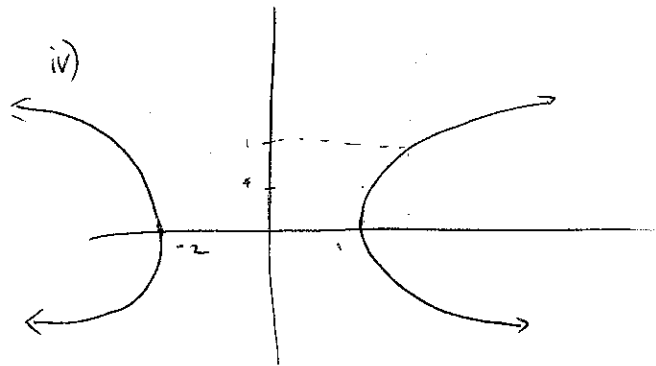
iii)



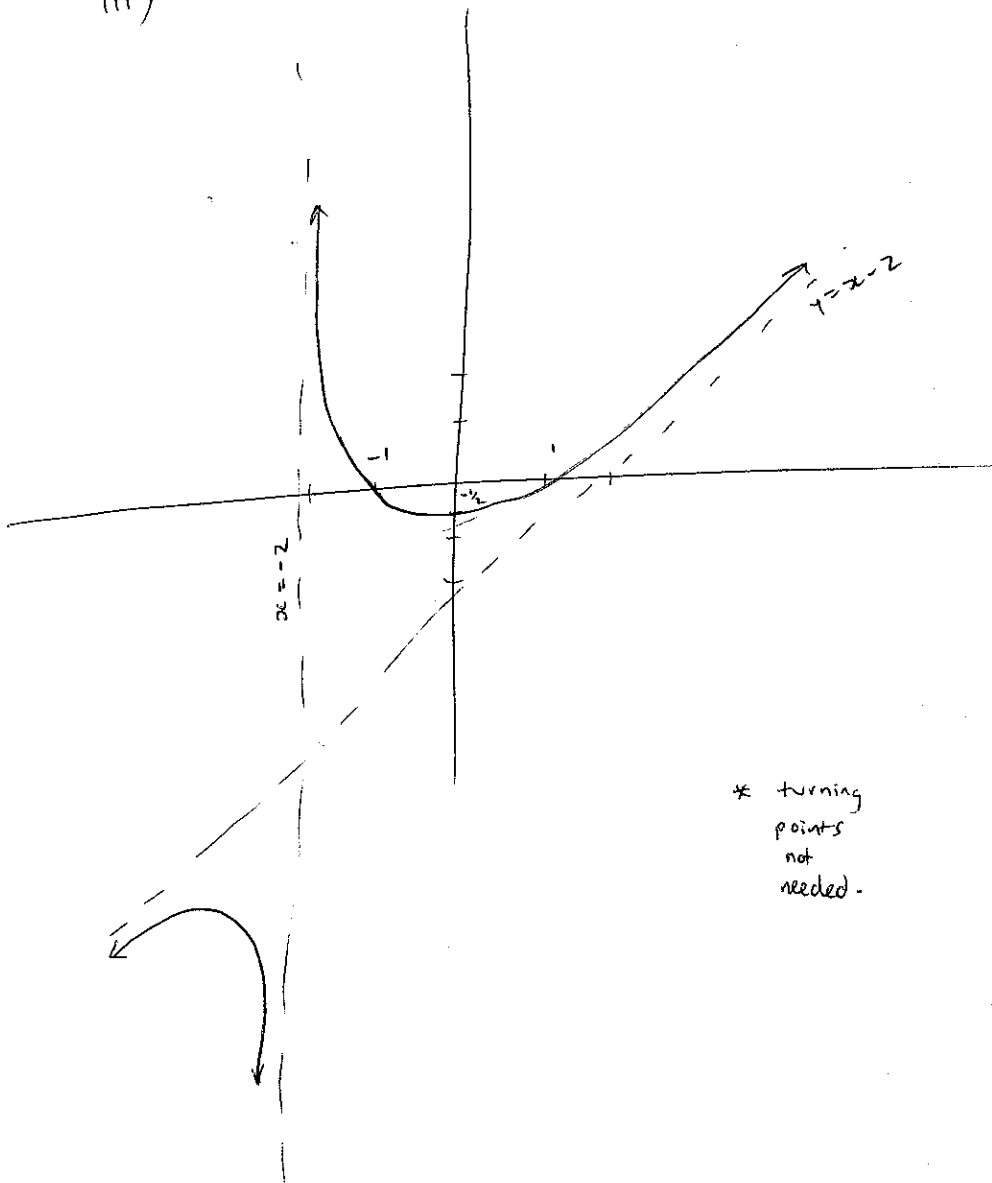
show x intercepts

$$(x+2)(x-1) = 1 \\ x^2 + 2x - x - 2 = 1 \\ x^2 + x - 3 = 0 \\ x = \frac{-1 \pm \sqrt{1+12}}{2} \\ x = \frac{-1 \pm \sqrt{13}}{2}$$

iv)



iii)



Q43

$$\begin{aligned} \text{a) } \int_0^{\pi/4} x \sin 2x \, dx &= \left[-\frac{x}{2} \cos 2x \right]_0^{\pi/4} \\ &+ \frac{1}{2} \int_0^{\pi/4} \cos 2x \, dx \\ &= 0 + \frac{1}{2} \left[\frac{\sin 2x}{2} \right]_0^{\pi/4} \\ &= \frac{1}{4} \end{aligned}$$

$$\begin{aligned} u &= x \\ u' &= 1 \\ v &= -\frac{1}{2} \cos 2x \\ v' &= \sin 2x \end{aligned}$$

$$uv - v u'$$

$$\begin{aligned} \text{b) } \int x \sqrt{1-x} \, dx & \quad \text{let } u = 1-x \\ du &= -dx \\ &= \int -(1-u)(u^{1/2}) \, du \\ &= \int -u^{1/2} + u^{3/2} \, du \\ &= -\frac{2}{3} u^{3/2} + \frac{2}{5} u^{5/2} + C \\ &= -\frac{2}{3} \sqrt{(1-x)^3} + \frac{2}{5} \sqrt{(1-x)^5} + C \end{aligned}$$

$$\begin{aligned} \text{let } u &= 1-x \\ du &= -dx \end{aligned}$$

$$\text{c) } \int \frac{1}{x(1+x^2)} \, dx = \frac{a}{x} + \frac{bx+c}{1+x^2}$$

$$\therefore 1 = a(1+x^2) + x(bx+c)$$

$$0 = a+b, \quad 0 = c, \quad 1 = a$$

$$\therefore b = -1$$

$$\therefore \int \frac{1}{x(1+x^2)} \, dx = \frac{1}{x} - \frac{x}{1+x^2}$$

$$= \ln x - \frac{1}{2} \ln(1+x^2) + C$$

$$\begin{aligned}
 d) \int \frac{dx}{\sqrt{x^2 - 4x + 1}} &= \int \frac{dx}{\sqrt{(x-2)^2 - 3}} \quad \left(\text{let } u = x-2 \right) \\
 &\quad \frac{du}{dx} = 1 \\
 &= \frac{du}{\sqrt{u^2 - 3}} \\
 &= \ln(u + \sqrt{u^2 - 3}) + C \quad (\text{from } \int \text{table}) \\
 &= \ln[x-2 + \sqrt{(x-2)^2 - 3}] + C
 \end{aligned}$$

e) The function does not exist at $x=3$, which is within the range of the integration.

$$\begin{aligned}
 f) \int_0^{\frac{\pi}{3}} \frac{\tan x}{1 + \cos x} dx &\quad , t = \tan \frac{x}{2} \\
 &\quad \frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2} \\
 &\quad \frac{dt}{dx} = \frac{1}{2} (1+t^2) \\
 &\quad dx = \frac{2}{1+t^2} dt \\
 &= \int_0^{\frac{\pi}{3}} \frac{\frac{2t}{1-t^2}}{1 + \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt \\
 &= \int_0^{\frac{\pi}{3}} \frac{4t}{1-t^2} dt \\
 &= \int_0^{\frac{\pi}{3}} \frac{4t}{1-t^2} dt = \int_0^{\frac{\pi}{3}} \frac{2t}{1-t^2} dt \\
 &= \left[-\ln(1-t^2) \right]_0^{\frac{\pi}{3}} = -\ln\left(\frac{3}{4}\right) = \ln \frac{4}{3}
 \end{aligned}$$

Q4

a) i) $x^2 + 4y^2 = 100$ or $\frac{x^2}{100} + \frac{y^2}{25} = 1$ $\therefore a=10, b=5$

$$b^2 = a^2(1-e^2)$$

$$25 = 100(1-e^2)$$

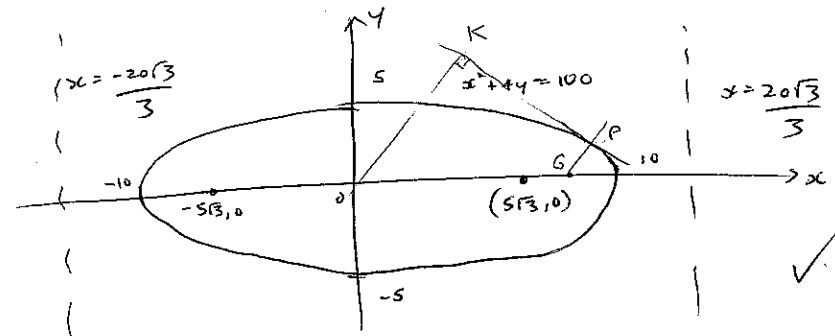
$$e = \frac{\sqrt{3}}{2}$$

$$\text{Foci} = (\pm ae, 0)$$

$$= (\pm 5\sqrt{3}, 0)$$

$$\text{Directrices: } \pm \frac{a}{e} = x = \pm \frac{20\sqrt{3}}{3}$$

ii)



iii) $x^2 + 4y^2 = 100$

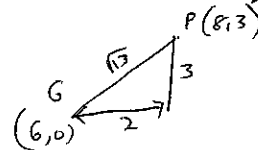
$$2x + 8y \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{4y} = -\frac{2}{3} \text{ at } P(8, 3)$$

$$\text{tangent: } y-3 = -\frac{2}{3}(x-8) \text{ or } 3y + 2x - 25 = 0$$

$$\text{normal: } y-3 = \frac{3}{2}(x-8) \text{ or } 3x - 2y - 18 = 0$$

iv) Normal at meets x -axis at $G(6, 0)$



$$\therefore PG = \sqrt{13}$$

$$OK = \frac{|-25|}{2^2 + 3^2} = \frac{25}{13}$$

$$\therefore PG \cdot OK = 25 = 5^2 \Rightarrow (\text{square of semi minor axis})$$

Qv4

$$b) V = \frac{u}{4+ux} \text{ ms}^{-1}$$

$$V^2 = \frac{u^2}{(4+ux)^2}$$

$$\frac{1}{2} V^2 = \frac{1}{2} u^2 (4+ux)^{-2}$$

$$\begin{aligned} \ddot{x} &= \frac{d}{dx} \left(\frac{1}{2} V^2 \right) = -u^3 (4+ux)^{-3} \\ &= \frac{-u^3}{(4+ux)^3} = -V^3 \text{ ms}^{-2} \end{aligned}$$

$$c) \frac{a(y)}{A} = \frac{(h-y)^2}{h^2}$$

$$\therefore a(y) = A \times \left(\frac{h-y}{h} \right)^2$$

$$ii) \Delta V = a(y) \Delta y$$

$$V = \frac{A}{h^2} \int_0^h (h-y)^2 dy$$

$$= \frac{-A}{3h^2} \left[(h-y)^3 \right]_0^h$$

$$= \frac{-A}{3h^2} (0-h^3)$$

$$= \frac{Ah}{3} = \frac{50000 \times 150}{3}$$

$$= 2500000 \text{ m}^3$$

Qv5

$$\begin{aligned} i) I_n &= \int_0^\pi x^n \sin x \, dx \\ &= \left[-x^n \cos x \right]_0^\pi + n \int_0^\pi \cos x \cdot x^{n-1} \, dx \\ &= \pi^n + n \left[\left[x^{n-1} \sin x \right]_0^\pi - \int_0^\pi (n-1)x^{n-2} \sin x \, dx \right] \\ &= \pi^n + n \left[0 - (n-1) I_{n-2} \right] \\ &= \pi^n - n(n-1) I_{n-2} \end{aligned}$$

Let $u = x^n$
 $\frac{du}{dx} = nx^{n-1}$
 $v = \sin x$
 $v = -\cos x$

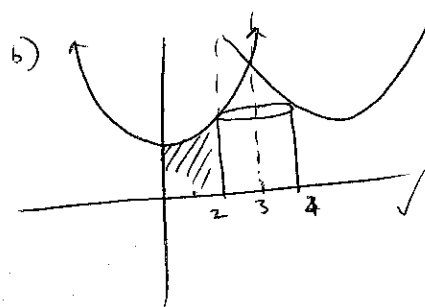
Let $u = x^{n-1}$
 $\frac{du}{dx} = (n-1)x^{n-2}$
 $v = \cos x$
 $v = \sin x$

$$ii) I_5 = \pi^5 - 5(4) I_3$$

$$I_3 = \pi^3 - 3(2) I_1$$

$$\begin{aligned} I_1 &= \int_0^\pi x \sin x \, dx \\ &= \left[x \cos x \right]_0^\pi + \int_0^\pi \cos x \, dx \\ &= \pi + \left[\sin x \right]_0^\pi \\ &= \pi \end{aligned}$$

$$\begin{aligned} \therefore I_5 &= \pi^5 - 20 \times [\pi^3 - 6(\pi)] \\ &= \pi^5 - 20\pi^3 + 120\pi \end{aligned}$$



$$V = 2\pi \int_0^2 (3-x)(x^2+1) \, dx$$

$$V = 2\pi \int_0^2 (3x^2 + 3 - x^3 - x) \, dx$$

$$V = 2\pi \left[x^3 + 3x - \frac{x^4}{4} - \frac{x^2}{2} \right]_0^2$$

$$V = 2\pi [8 + 6 - 4 - 2]$$

$$V = 16\pi \text{ u}^3$$

Q5

c) by de Moivre

$$(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta$$

$$(\cos \theta + i \sin \theta)^3 = \cos^3 \theta + 3\cos^2 \theta \cdot i \sin \theta + 3\cos \theta \cdot i^2 \sin^2 \theta + i^3 \sin^3 \theta \quad \checkmark$$

$$= \cos^3 \theta - 3\cos \theta \sin^2 \theta + i(3\cos^2 \theta \sin \theta - \sin^3 \theta) \quad \checkmark$$

Equating Real

$$\cos 3\theta = \cos^3 \theta - 3\cos \theta \sin^2 \theta \quad \checkmark$$

ii) Equating imaginary

$$\sin 3\theta = 3\cos^2 \theta \sin \theta - \sin^3 \theta \quad \checkmark$$

Q6

a) i) $\frac{1}{x}, \frac{1}{x}, \frac{1}{x}$ will satisfy

$$\left(\frac{1}{x}\right)^3 + q\left(\frac{1}{x}\right) + r = 0 \quad \checkmark$$

$$1 + qx^2 + rx^3 = 0 \quad \checkmark$$

ii) x^2, β^2, γ^2 will satisfy

$$(\sqrt{x})^3 + q(\sqrt{x}) + r = 0$$

$$\sqrt{x}(x + q) = -r \quad \checkmark$$

$$x(x + q)^2 = r^2$$

$$x(x^2 + 2xq + q^2) - r^2 = 0$$

$$x^3 + 2x^2q + xq^2 - r^2 = 0 \quad \checkmark$$

b) I) true, $P(x)$ will be in form

$$P(x) = ax^{2n} \dots$$

$$P'(x) = 2nax^{2n-1} \dots$$

leaving odd powers

II) false, due to constant term.
all terms odd except constant term.

Q6

c) i) $\int_0^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^2}} = \left[\sin^{-1} x \right]_0^{\frac{1}{2}} = \sin^{-1} \frac{1}{2} = \frac{\pi}{6}$ ✓

ii) For $n \geq 2$ and $0 \leq x \leq \frac{1}{2}$

$0 \leq x^n \leq x^2$ ✓

$1 \geq 1-x^n \geq 1-x^2$ ✓

$1 \geq \sqrt{1-x^n} \geq \sqrt{1-x^2}$ ✓

iii) $1 \leq \frac{1}{\sqrt{1-x^n}} \leq \frac{1}{\sqrt{1-x^2}}$ ✓

$\int_0^{\frac{1}{2}} 1 dx \leq \int_0^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^n}} \leq \int_0^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^2}}$ ✓

* $\frac{1}{2} \leq \int_0^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^n}} \leq \frac{\pi}{6}$ ✓

$1 > x^n > 0$

$1-x^n > 0$

$\therefore 1 > 1-x^n$

$x^2 > x^n$

$(\text{since } x \leq \frac{1}{2})$

$\therefore 1-x^2 > 1-x^n$

70

Step 1

Prove true when $n=1$ and $n=2$

when $n=1$, $a_1 = 2^{1-1} + 1 = 2$

when $n=2$, $a_2 = 2^{2-1} + 1 = 3$

$\therefore$ true for $n=1$ and $n=2$

Step 2

Assume true for $n=k$ and when $n=k-1$.

ie, $a_k = 2^{k-1} + 1$ (*) and $a_{k-1} = 2^{k-2} + 1$ (**)

Step 3

Prove true for $n=k+1$

ie, $a_{k+1} = 2^k + 1$ required to prove.

As $a_n = 3a_{n-1} - 2a_{n-2}$

$\therefore a_{k+1} = 3a_k - 2a_{k-1}$ ✓

$= 3(2^{k-1} + 1) - 2(2^{k-2} + 1)$ from assumption (*)

$= 3 \times 2^{k-1} + 3 - 2 \times 2^{k-2} - 2$ ✓

$= 3 \times 2^{k-1} - 2^{k-1} + 1$ ✓

$= 2 \times 2^{k-1} + 1$ ✓

$= 2^k + 1$ ✓

$\therefore$ result true for $n=k+1$

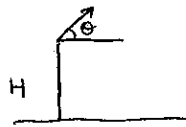
Step 4 : conclusion.

5/10/21

1/2/22

1/2/22

Q7) b) i)



$$\ddot{x} = 0$$

$$\dot{x} = v \cos \theta$$

$$x = vt \cos \theta$$

$$\ddot{y} = -g$$

$$\dot{y} = v \sin \theta - gt$$

$$y = vt \sin \theta - \frac{1}{2}gt^2 + H$$

✓ [for equations]

when $y = -H$

$$\frac{1}{2}gt^2 - vt \sin \theta - H = 0$$

$$t = \frac{v \sin \theta \pm \sqrt{v^2 \sin^2 \theta + 2gH}}{g}$$

(disregard -)

$$\therefore R = \frac{v \cos \theta (v \sin \theta + \sqrt{v^2 \sin^2 \theta + 2gH})}{g}$$

✓ (sub t into x)

$$\frac{gR}{v \cos \theta} = v \sin \theta + \sqrt{v^2 \sin^2 \theta + 2gH}$$

$$\sqrt{v^2 \sin^2 \theta + 2gH} = \frac{gR}{v \cos \theta} - v \sin \theta$$

$$v^2 \sin^2 \theta + 2gH = \frac{g^2 R^2}{v^2 \cos^2 \theta} - \frac{2gR \sin \theta}{\cos \theta} + v^2 \sin^2 \theta$$

$$2gH = \frac{g^2 R^2 \sec^2 \theta}{v^2} - 2gR \tan \theta$$

$$g^2 R^2 \sec^2 \theta = 2v^2 gH + 2gR \tan \theta v^2$$

$$gR^2 \sec^2 \theta = 2v^2 (H + R \tan \theta) \quad - (1)$$

~~1/11/11~~

ii)

$$2v^2(R \tan \theta + H) = gR^2 \sec^2 \theta$$

$$2v^2 \left(R \sec^2 \theta + \tan \theta \cdot R \frac{d}{d\theta} \right) = gR^2 2 \sec^2 \theta \tan \theta + g \sec^2 \theta \cdot R^2 \frac{d}{d\theta}$$

Note: $\left[\frac{d}{d\theta} (\cos \theta)^2 = -2(\cos \theta)^3 \cdot \sin \theta \right]$
 $= 2 \sec^2 \theta \tan \theta$

$$2v^2 \left(R \sec^2 \theta + \tan \theta \frac{dR}{d\theta} \right) = gR^2 2 \sec^2 \theta \tan \theta + g \sec^2 \theta 2R \frac{dR}{d\theta}$$

$$\frac{dR}{d\theta} (2v^2 \tan \theta - 2gR \sec^2 \theta) = 2gR^2 \sec^2 \theta (gR \tan \theta - v^2)$$

$$\therefore \frac{dR}{d\theta} = 0 \text{ when } gR \tan \theta - v^2 = 0 \text{ or } \boxed{\tan \theta = \frac{v^2}{gR}} \quad - (2)$$

sub (2) in (1) for max value

$$2v^2 \left(\frac{v^2}{g} + H \right) = gR^2 (1 + \tan^2 \theta)$$

$$= gR^2 + gR^2 \left[\frac{v^2}{gR} \right]^2 = \cancel{1/11/11}$$

$$2v^2 (v^2 + gH) = g^2 R^2 + v^4$$

$$2v^4 + 2v^2 gH = g^2 R^2 + v^4$$

$$g^2 R^2 = v^4 + 2v^2 gH$$

$$R^2 = \frac{v^4}{g^2} + \frac{2v^2 gH}{g^2}$$

$$R^2 = \frac{v^2}{g^2} (v^2 + 2gH) \quad (*) \quad \theta_1 = \tan^{-1} \left(\frac{v^2}{gR} \right)$$

$$R_1 = \frac{v}{g} \sqrt{v^2 + 2gH}$$

$$\text{iii) } \tan 2\theta_1 = \frac{2 \tan \theta_1}{1 - \tan^2 \theta_1} = \frac{2 \cdot \left(\frac{v^2}{gR_1} \right)}{1 - \left(\frac{v^2}{gR_1} \right)^2} = \frac{2gR_1 v^2}{g^2 R_1^2 - v^4}$$

$$= \frac{2gR_1 v^2}{2v^2 gH} = \frac{R_1}{H}$$

from (2)

Q8

a) $m = \text{cis } \frac{2\pi}{7}$

$m^7 = (\text{cis } \frac{2\pi}{7})^7$

$m^7 = \text{cis } 2\pi = 1$

$\therefore m$ is a root of $x^7 - 1 = 0$

$\therefore (x-1)(x^6 + x^5 + x^4 + x^3 + x^2 + x + 1) = 0$

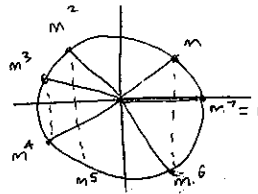
Since $m \neq 1$

m satisfies $x^6 + x^5 + x^4 + \dots + x + 1 = 0$

$\therefore 1 + m + m^2 + \dots + m^6 = 0$

ii) since coefficients of $x^2 + ax + b = 0$ are real and $\alpha = m + m^2 + m^4$ is a complex root, then $\bar{\alpha}$ is also a root.

$\therefore \beta = \bar{\alpha} = \overline{m + m^2 + m^4}$
 $= \bar{m} + \bar{m}^2 + \bar{m}^4$
 $= m^6 + m^5 + m^3$



iii) $\alpha + \beta = -a$

$a = -(\alpha + \beta)$

$a = -(m + m^2 + m^3 + m^4 + m^5 + m^6)$ from ii)

$a = 1$

$\alpha\beta = b$
 $= (m + m^2 + m^4)(m^3 + m^5 + m^6)$
 $= m^4(1 + m + m^3)(1 + m^2 + m^3)$
 $= m^4(1 + m^2 + m^3 + m + m^3 + m^4 + m^3 + m^5 + m^6)$
 $= m^4(1 + m + m^2 + m^3 + m^4 + m^5 + m^6 + 2m^3)$
 $= m^4(0 + 2m^3)$
 $= 2m^7$

$b = 2$

iv) from iii) $x^2 + x + 2 = 0$

$x = \frac{-1 \pm \sqrt{1-8}}{2}$

$x = \frac{-1 \pm i\sqrt{7}}{2}$

$\therefore \text{Im}(x) = \pm \frac{\sqrt{7}}{2}$

However, $\alpha = m + m^2 + m^4$
 + from ii) diagram $\text{Im}(m + m^2 + m^4) > 0$

$\therefore \text{Im}(\alpha) = \frac{\sqrt{7}}{2}$

Now, $\text{Im}(\alpha) = \text{Im}(m + m^2 + m^4)$
 $= \sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7}$
 $= \sin \frac{2\pi}{7} + \sin \frac{3\pi}{7} - \sin \frac{\pi}{7}$
 $= \frac{\sqrt{7}}{2}$ as required.

Q8 b)

i) $1 \geq 3\sqrt[3]{abc}$

$\frac{1}{3} \geq \sqrt[3]{abc}$

$\therefore \sqrt[3]{abc} \geq \frac{1}{3}$

Using 2nd given result $a+b+c \geq 3\sqrt[3]{abc}$

$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 3\sqrt[3]{\frac{1}{abc}}$

$= 3\sqrt[3]{abc}$

$\geq 3 \cdot 3$

≥ 9

ii) consider $(\frac{1}{a}-1)(\frac{1}{b}-1)(\frac{1}{c}-1)$

$= \frac{1-a}{a} \times \frac{1-b}{b} \times \frac{1-c}{c}$

$= \frac{b+c}{a} \times \frac{a+c}{b} \times \frac{a+b}{c}$

$\geq \frac{2\sqrt{bc} \cdot 2\sqrt{ac} \cdot 2\sqrt{ab}}{abc}$

≥ 8

$b+c \geq 2\sqrt{bc}$
 $a+c \geq 2\sqrt{ac}$
 $a+b \geq 2\sqrt{ab}$

[note = 8 when $a=b=c = \frac{1}{3}$]